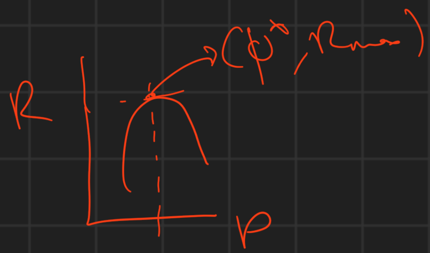
W5

→ Optimization → Primal Dual Conversion

(1) Revenue :

Revenue = SP * QE.

$R(p) = p * D(p)$



if $D(p) = D_0 - mp$

$R(p) = (D_0 - mp)(p) = D_0p - mp^2$

$\frac{dR(p)}{dp} = D_0 - 2mp = 0 \Rightarrow p^* = \frac{D_0}{2m}$ [F.O.C]

$\frac{d^2R}{dp^2} = -2m < 0$ [S.O.C]

$R_{Max} (p = D_0/2m)$

(2) Profit

$\pi(p) = \text{Revenue} - TC = (p - c)(D(p))$

$\pi(p) = (p - c)(D_0 - mp)$

$\frac{d\pi}{dp} = 0$ [F.O.C] $\Rightarrow p^* = \frac{D_0 + mc}{2m}$

Primal Dual Conversion

LPP

→ Define Decision Var.
→ Define Constraints
→ Define Objective func

Max / Min $\sum a_i x_i$

s.t. $\sum b_{ij} x_j \leq 0$

⋮

Max Z	Min Z
s.t. $\leq$	s.t. $\geq$
$<$	$>$
✗ $\{ >$ ✗, $\geq$ ✗	✗ $\{ <$ ✗, $\leq$ ✗

$=$?

✓ $<=$

$>=$ ✗ → $\times -1$ $<=$

Binding Constraint
→ optimal values give LHS = RHS
→ Corresponding Dec. Var is Non Zero

- Primal-Dual Conversion:

Primal	Dual
Maximization (or Minimization)	Minimization(or Maximization)
Number of Constraints	Number of Decision Variables
Number of Decision Variables	Number of Constraints
Objective Func. Coefficient	RHS of Constraint
RHS of Constraint	Objective Func. Coefficient

Opposite for Non Binding :

LHS ≠ RHS

(Ex)

Max: $a_1 x_1 + a_2 x_2 + a_3 x_3$

St:

$$c_1 x_1 + c_2 x_2 \leq d$$

$$p_1 x_1 + p_2 x_3 \leq e$$

Min $d\, y_1 + e\, y_2$

$$\_\ y_1 + \_\ y_2 \geq a_1$$

$$\_\ y_1 + \_\ y_2 \geq a_2$$

$$\_\ y_1 + \_\ y_2 \geq a_3$$

PYQ

Q1)

"Fake Money (FM)" is an investment consulting firm.
Dr. Milo has approached FM with Rs. 1,00,000 for guidance on investment opportunities. Dr. Milo wants to invest the entire Rs. 1,00,000. FM's top financial analyst recommends that all new investments be made in the oil industry, steel industry, or government bonds. Specifically, the analyst identified five investment opportunities and projected their annual rates of returns (which are specified in Table 1). Dr. Milo is a guy who likes to distribute his investments.

Hence, he has specified the following requirements to the analyst:
1. Neither industry (oil or steel) should receive more than Rs. 50,000
2. Government bonds should be at least 25% of the steel industry investments
3. The investments in Bongu Oil, the high-return but high-risk investment, cannot be more than 60% of the total oil industry investment

	Investment Opportunity	*Projected Annual Rate of Return (in %)*
x_1	Dhokha Oil	7.3
x_2	Bongu Oil	10.3
.	Achchha Steel	6.4
:	Nalla Steel	7.5
x_5	Government Bonds	4.5

Q1) How many decision variable(s) is/are present in the standard form of the primal?

Q2) How many constraints (excluding the non-negativity constraints) are present in the standard form of the primal?

Q3) How many constraints (excluding the non-negativity constraints) are present in the dual (which is based on the standard form of the primal)?

Q4) If FM has suggested the below distribution of investments, then how much money (total money) is projected to be in hand at the end of the first year? (Note: Enter the answer in "Rs." rounded to two decimal places without the "Rs." symbol. For example, if the answer is "Rs. 1.234", then enter it as "1.23")

Q5) If FM has suggested the below distribution of investments, then how many decision variables in the dual will have a non-zero value?

Investment Opportunity	*Amount Invested (in Rs.)*
Dhokha Oil	20,000
Bongu Oil	30,000
Achchha Steel	0
Nalla Steel	40,000
Government Bonds	10,000

(a) 5 DV $\rightarrow$ x_1 ...... x_5

(b) $\sum^{5} x_i = 1,00,000$ ....

$x_1 + x_2 <= 50000$

$$x_5 >= 0.25(x_3 + x_4) \rightarrow -x_5 <= -0.25(x_3+x_4)$$

$$x_2 <= 0.6(x_1 + x_2)$$

$$\sum x_i = 1L \quad \begin{cases} \sum x_i <= 1L \\ \sum x_i >= 1L \\ -\sum x_i <= -1L \end{cases}$$

$\boxed{6 \text{ constraint}}$

100×1.06

$100 \xrightarrow{6\%} \boxed{106}$ (6)

(d)

$20K \times 1.073 + \cdots\cdots\cdots \; 10K \times 1.045$

$=$ 108000

or (8000 interest)

(e)

Investment Opportunity	Amount Invested (in Rs.)
Dhokha Oil	20,000
Bongu Oil	30,000
Achchha Steel	0
Nalla Steel	40,000
Government Bonds	10,000

(1) $\sum x_i <= 1L$ ✓ Bindin

(2) $-\sum x_i <= -1L$ ✓ Binding

(3) $x_1 + x_2 <= 50000$ ✓ Binding

(4) $x_3 + x_4 <= 80K$

$40K \neq 80K$ ✗ Not Bind

(5) $x_5 >= 0.25(x_3 + x_4)$

(6) $x_2 <= 0.6(x_1 + x_2)$

(Q2)

The demand for a product is estimated to follow the relationship: D(p) = 1444 – 3(p^2). Here, "p" is the selling price for the product. The product is made at Rs. 50 per unit. Given this information, answer the following questions.

a) If you intend to maximize the revenue, then what is the optimal value for "p"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

b) If you intend to maximize the profits, then what is the optimal value for "p"? (Note: Enter the answer rounded to two decimal places. For example, if the answer is "1.234", then enter it as "1.23")

$$D(p) = 1444 - 3p^2$$

$$R(p) = D(p) \times p = 1444p - 3p^3$$

$$\frac{\partial R}{\partial p} = 0 \Rightarrow 1444 - 9p^2 = 0 \rightarrow p^2 = \frac{1444}{9}$$

$$\boxed{p = 12.666..}$$

(b) $\pi(p) = (D(p)) * (p - c)$

$$= 1444 - 3p^2 * (p - 50)$$

$$= -3p^3 + 150p^2 + 1444p - 72200$$

$$\frac{d\pi}{dp} = 0 \Rightarrow -9p^2 + 300p + 1444 = 0$$

$$\boxed{p_1 = -4.28} \quad \& \quad \boxed{p_2 = 37.60}$$

$D(p = 37.60) = -ve$

(Q3)

Say a demand response curve is modeled as a constant elasticity curve. If Q1 is 2400 units, Q2 is 1500 units, P1 is Rs. 100 and P2 is Rs. 200, then what is the elasticity of the curve? (Note: Enter the answer rounded to two decimal places. For example, if the answer is “1.234”, then enter it as “1.23”)

$$D(p) = C \cdot p^{-\varepsilon}$$

$$2400 = C \cdot 100^{-\varepsilon} \quad \text{— (1)}$$

$$1500 = C \cdot 200^{-\varepsilon} \quad \text{— (2)}$$

$$(1) \div (2)$$

$$\left(\frac{2400}{1500}\right) = \left(\frac{100}{200}\right)^{-\varepsilon} \Rightarrow \varepsilon = \frac{\ln(D_2/D_1)}{\ln(P_1/P_2)} = \frac{\ln(D_1/D_2)}{\ln(P_2/P_1)}$$

$$= \frac{\ln(24/15)}{\ln(200/100)} = \boxed{0.6780}$$

Q4)

The primal formulation of a production planning problem, where the objective is to maximize the total profit is formulated as specified below.

Maximize: 225*X1 + 200*X2 + 165*X3

Subject to the following constraints

→ Bind

13*X1 + 26*X2 + 71*X3 <= 100012	Constraint-1
45*X1 + 10*X2 + 5*X3 <= 100000	Constraint-2
12*X1 + 15*X2 + 34*X3 <= 100000	Constraint-3
0*X1 + 45*X2 + 4*X3 <= 100000	Constraint-4
X1 >=0	Constraint-5
X2 >=0	Constraint-6
X3 >=0	Constraint-7

<= , <

→ Non-B

a) How many constraints will be present in the dual formulation (after converting the primal to the standard form)?

b) How many decision variables will be present in the dual formulation? (after converting the primal to the standard form)

c) If the optimal solution of the primal is X1= 1701, X2= 2196, X3= 293. Then, how many decision variables will have a value of "0" in the optimal solution of the dual (after converting the primal to

3

(c) $13(1701) + 26(2196) + 71(293)$

$= 100012 = RHS$

1 const → Binding (LHS = RHS)

3 const → Non Binding (LHS ≠ RHS)

→ {DV corresponding to constraint where LHS ≠ RHS}

(Q5)

Milo The Goofball (MTG) is a company that makes footballs for “Recreation (R)” games and “Professional (P)” games. The process of making a football involves the following three stages
Stage-1: Cutting and Dyeing
Stage-2: Sewing
Stage-3: Inspection

It is seen that making a R-game football requires 6 minutes in Stage-1, 15 minutes in Stage-2 and 6 minutes in Stage-3. On the other hand, the football for a P-game requires 12 minutes, 9 minutes and 6 minutes in Stages 1,2 and 3 respectively.

For the coming planning period of 4 weeks, MTG has 340 hours available in Stage-1, 480 hours available in Stage-2 and 300 hours available in Stage-3. Moreover, if required, MTG can avail more hours in Stage-1 at an additional cost of Rs. 200 per hour.

Moreover, a R-game football generates a profit of Rs. 500 per piece and a P-game football generates a profit of Rs. 800 per piece. Furthermore, MTG has forecasted that the maximum number of footballs it can sell is 3000. With an objective to maximize the profit, formulate the problem as a Linear Program and answer the given subquestions.

DV → How many P type & R type football we need

$\boxed{n_p, \; n_r \quad t_1}$

a) How many decision variables are present in the standard form of the primal linear Programming Problem?

b) How many decision variables are present in the Dual Linear Programming Problem (which is developed based on the standard form of the problem)?

c) If the company decides to make 1500 P-game footballs, 1000 R-game footballs, then how many decision variables in the Dual Linear Programming Problem (which is developed based on the standard form of the problem) will have a value of “0”?

d) If the company decides to make 1500 P-game footballs, 1000 R-game footballs, then what is the objective function value for the primal linear programming problem (which is in the standard form)? [Note: Enter your answer rounded to two decimal places. For example, if your answer is 1.235 then enter the answer as “1.24”]

e) If the MTG decides to make 1500 P-game footballs, 1000 R-game footballs, then which of the following options is/are correct

Options :
- MTG will purchase addition capacity in Stage-1
- MTG will not purchase addition capacity in Stage-1
- MTG will have excess capacity in Stage-1
- MTG will have excess capacity in Stage-2
- MTG will have excess capacity in Stage-3

$$\text{Max } \left(\underbrace{N_p 800 + N_R 500 - 200 t_1}_{} \right)$$

st:

$$6N_R + 12N_p \quad <= \quad (340 \times 60)\text{ min}$$

$$15 N_R + 9 N_P \quad <= \quad (480 \times 60)$$

$N_p + N_k \leq 3000$

$N_p = 1500$, $N_f = 1000$

① 24000 | $\boxed{20400}$ (f)

We need 60 hr more in Stg 2.

② 20500 | 20800 (f)

③ 13000 | 12000 (f)

2000 f 3000 (f)

$1500 \times 800 + 1000 \times 500 - 200 \times 60$

$= 16,88,000$

d(i)

You solve the primal of a linear program with a maximization objective, three decision variables and two constraints of the less than or equal to type. Non-negativity restrictions apply to the decision variables. After solving the linear program, you find that the first constraint is not binding (LHS < RHS) and the second constraint is binding (LHS = RHS). Which of the following statements is/are correct?

a) There are three decision variables in the dual
b) The dual variable corresponding to the second constraint is zero
c) There are two decision variables in the dual formulation
d) The dual variable corresponding to the second constraint is non-zero